



THE SHINE STAR EXAMINATIONS BOARD
PRE - PLE SET FOUR (PINNACLE)

2025

MATHEMATICS

Time Allowed: 2 hours 30 minutes

Random No.					Personal No.		

Candidate's Name:

Candidate's Signature:

District ID No.

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Read the following instructions carefully;

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two sections: **A** and **B**. Section **A** has **20** questions and section **B** has **12** questions
3. Answer **all** the questions. **All** the answers to both sections **A** and **B** must be written in the spaces provided.
4. **All** the answers **must** be written using a **blue** or **black** ball point pen or ink. Any work written in pencil will **not** be marked.
5. **No calculators** are allowed in the examination room.
6. Unnecessary **changes** in your work and handwriting that cannot be read easily may lead to **loss of marks**.
7. Do not fill anything in the table indicated:
"FOR EXAMINERS' USE ONLY" and boxes inside the question paper.

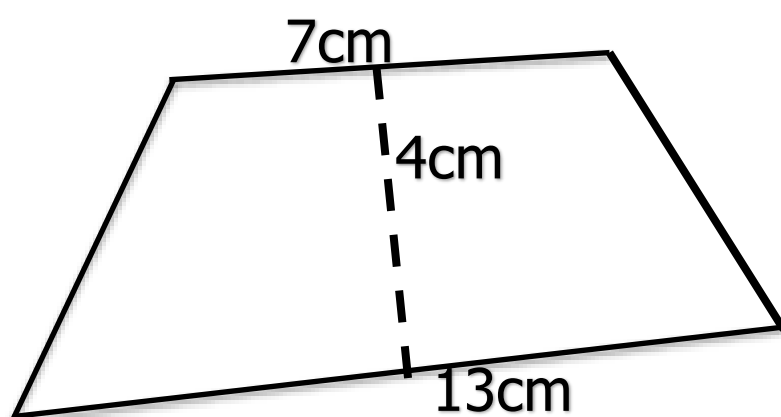
FOR EXAMINERS USE ONLY		
QN NO	MARKS	EXR'S NO.
1-5		
6-10		
11-15		
16-20		
21-22		
23-24		
25-26		
27-28		
29-30		
31-32		
TOTAL		

Turn over

SECTION A: 40 MARKS

Question 1-20, carry two marks each.

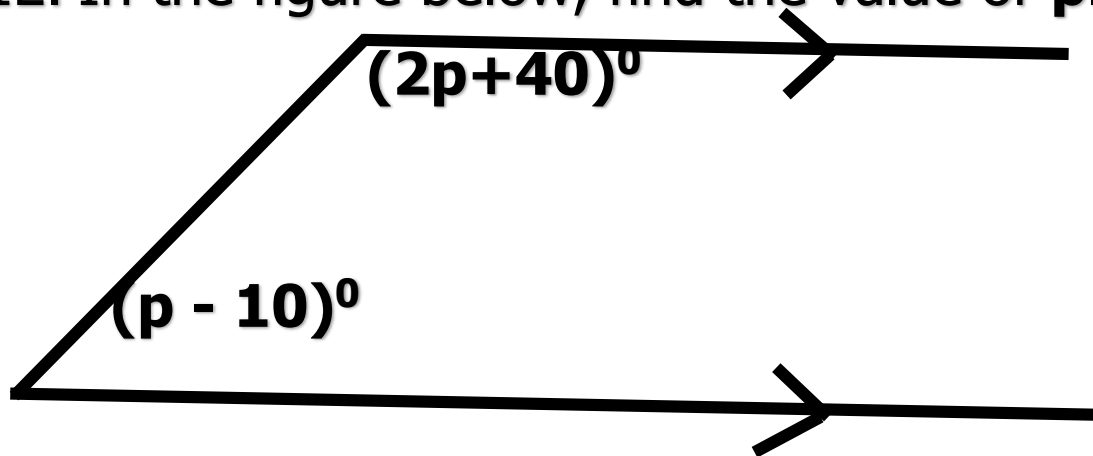
1. Subtract: $46-28$.
2. Write 1101121 in words.
3. Express 2:5 as a common fraction.
4. Given that set $Q =$ (the first six triangular numbers).Find the sum of last two numbers in set Q .
5. The figure below is an isosceles trapezium, use it to find its perimeter.



6. Work out $330_{\text{four}} - 23_{\text{four}}$.
7. Shamin bought 4 exercise books at sh. 1600. Find the cost of 2 similar books.
8. Simplify : $2(\frac{1}{2}k+6)$
9. Express 12:45pm as a 24 hour clock.
10. Find the complements of $(k+20)^0$

11. The sum of interior angles of a regular polygon is 12 right angles.
Name the polygon.

12. In the figure below, find the value of **p**.



13. What is 12% of 2000balls?

14. In a class of girls and boys, the probability of picking a boy to become a class monitor is $\frac{4}{9}$. If there are 30 girls in the class, how many more boys than girls are in the class?

15. Work out $(0.2)^2 \times 25$

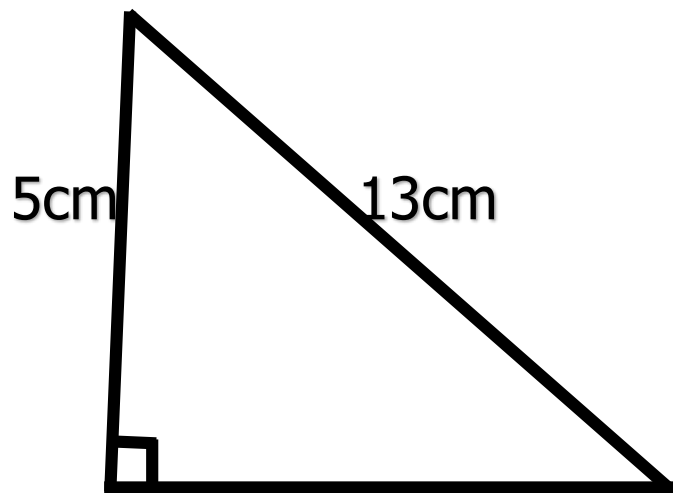
16. How many lines of folding symmetry has an isosceles triangle (show them).

17. Nyeko's father takes only 30minutes to drive him to school in the morning. What time must he leave home in order to get to school at 7:30am?

18. Subtract 0.07 from 1.

19. If $\frac{t}{2} = \frac{32}{t}$, find the value of t.

20. Calculate the area of the triangle ABC. Below



SECTION B: 60 MARKS

Question 21-32, carry marks in brackets

21. Simplify:

a) $1.4 - 3.6 + 4.0$

(2mrks)

b) Work out : $\frac{0.16 \times 0.24}{0.004}$

(3mrks)

22. Busega p/s bought some land for its staff both teaching and non-teaching staff who are 49 members in total. If 20 hectares of land were set aside for future development while each member was allocated a hectare, how much land did the school buy?

(3mrks)

(b). The range of two numbers is **15**. If the smaller number is **97**. Find the other number.

(2mrks)

23. Given that set $Q = \{\text{all prime numbers less than } 10\}$
 $R = \{\text{all odd numbers less than } 10\}$

Find:

i) Members of set R.

(2mrks)

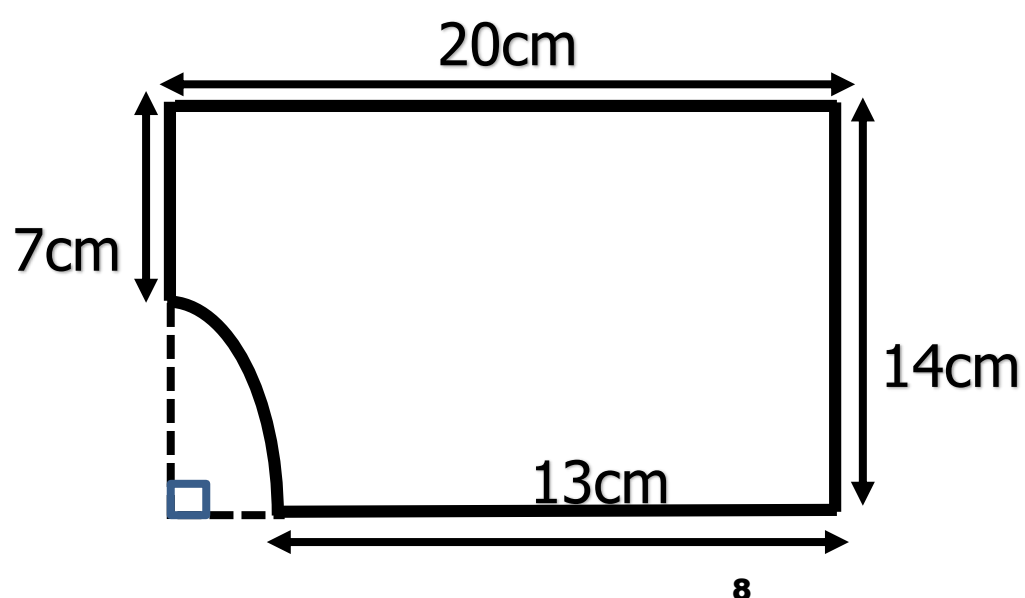
ii) $n(Q-R)$

(2mrks)

iii) List down all the subsets of $(Q \cap R)$

(1mrk)

24. Study the figure below and use it to answer the questions that follow.



(a) Find the distance round the figure above.

(3mrks)

(b) Find the area of the above figure.

(2mrks)

25. Express 403_{five} to base eight

(3mks)

b) Find the value of t where $41_t = 11001_{\text{two}}$

(3mks)

26. Gladys left town T and travelled westwards to village V at an average speed of 50km/h for 2hours from the village, he drove southwards to market M at 25km/h for 2hours.

a) Draw a rough sketch to show Gladys' journey.

(2mks)

b) Using a scale of 1cm to represent 20km, construct an accurate diagram to show Gladys' journey.

(3mks)

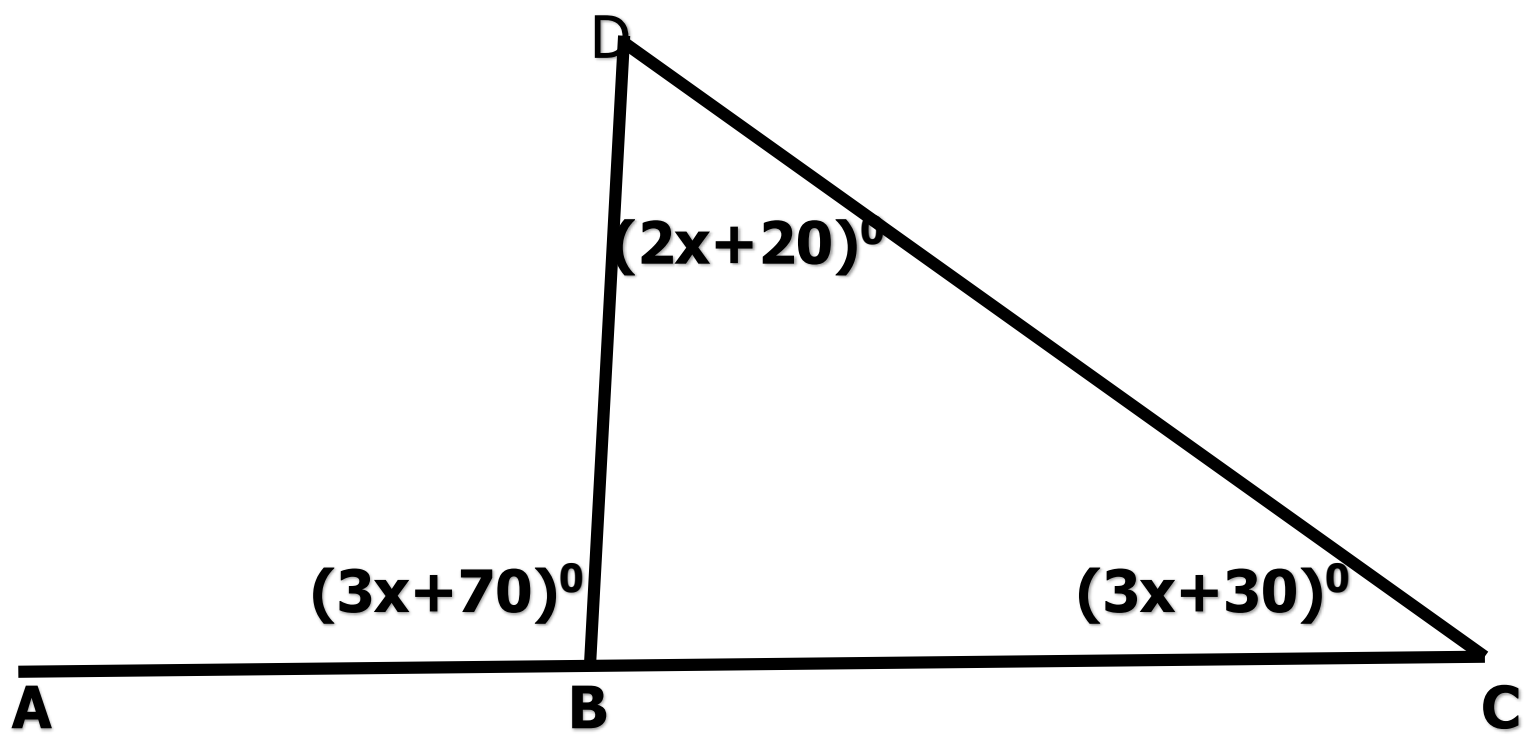
27. Mukeba was born in 15BC and he died in 35AD after his birth day.

a) How old was he when he died? *(2mrks)*

b) Musolin had a debt of sh.13,000 from each of his 4 friends .He received sh.50,000 as salary. How much money was Musolin left with after paying off sh.50,000?

(2mks)

28. In the diagram below, ABC is a straight line, BDC is a triangle. Use the given information to answer the questions that follow.



a) Find the value of x .

(2mks)

b) Find the size of angle BDC

(2mks)

29. Bazibu drove from town P to town Q at an average speed of 30km/h for 2hours.He continued to town R driving at a steady speed of 60km/h for 1hour.He the continued to town P at a steady speed of 49km/h. If he left town P at 8:00am.

a) At what time did he reach his final destination?

(2mrks)

b) Calculate the average speed for the whole journey.

(3mrks)

30. Kiberu went to the market for shopping with sh. 50000 to buy the items as shown in the table below.

Item	Quality	Unit cost	Total cost
Meat	2kg	sh.8000	sh. _____
Rice	_____kg	sh2500	sh. 10000
Sugar	$1\frac{1}{2}$ kg sh.	Sh.4500	Sh _____
Onion	500g	sh.3000	sh. _____
Total Expenditure			sh. _____

a) Complete the table above.

(5mrks)

b) If he was given a discount of 20% on his total Expenditure, how much was the discount.

(1mk)

31. Solve for:

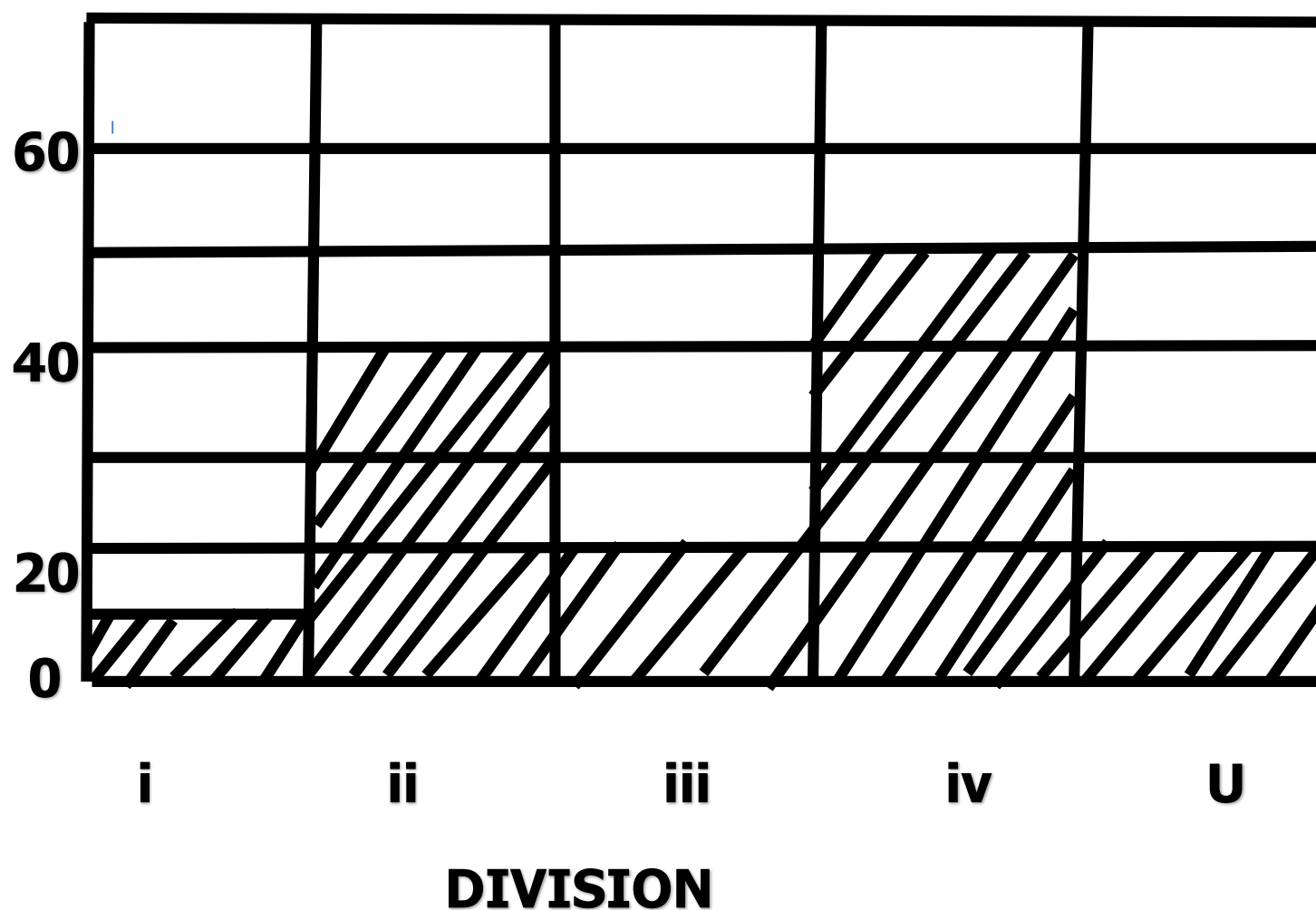
a) **P:** $\frac{2}{5}P + 3 \geq 5$

(2mrks)

b) Namuli bought 3 exercise books each at sh.(x+50) and 4 pencils each at sh.(x-50).If her total expenditure was sh.650,how much did she spend on pencils?

(2mks)

32. The graph below shows the number of candidates who sat their PLE 2024 at Grace of God P/S. They were 140 candidates altogether. Study the graph and answer the questions that follow



a) How many candidates passed in first grade?

(1mk)

b) How many candidates passed that year?

(2mrks)

c) How many more candidates passed in Fourth grade than those who failed?

(2mrks)